



FutureProof — Equity Preservation Mortgage

Wholesale Funding Partnership Proposal

Credit Risk Analysis & Portfolio Economics | February 2025

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Executive Summary

FutureProof seeks a wholesale funding partnership to originate Equity Preservation Mortgages — a new class of residential mortgage product with unique risk characteristics. This document presents independent quantitative analysis demonstrating that EPMs, when structured with Principal & Interest repayment and buffer ETF allocation, exhibit loss characteristics comparable to or better than prime residential mortgages.

Key Points for Wholesale Funders

- **Ultra-low default risk** — P&I + Buffer ETF structure achieves 0.5% deficit probability over 30 years (50K-path Monte Carlo validated)
- **Fully collateralised** — Secured against residential property (\$2M+ values) with 60-80% LVR
- **Self-liquidating** — P&I structure amortises loan to zero over tenure; no balloon risk
- **Equity buffer** — Investment portfolio provides additional collateral above and beyond the property
- **Wholesale margin sensitivity** — Product viability improves significantly with competitive funding; a partnership approach creates a better product and larger addressable market

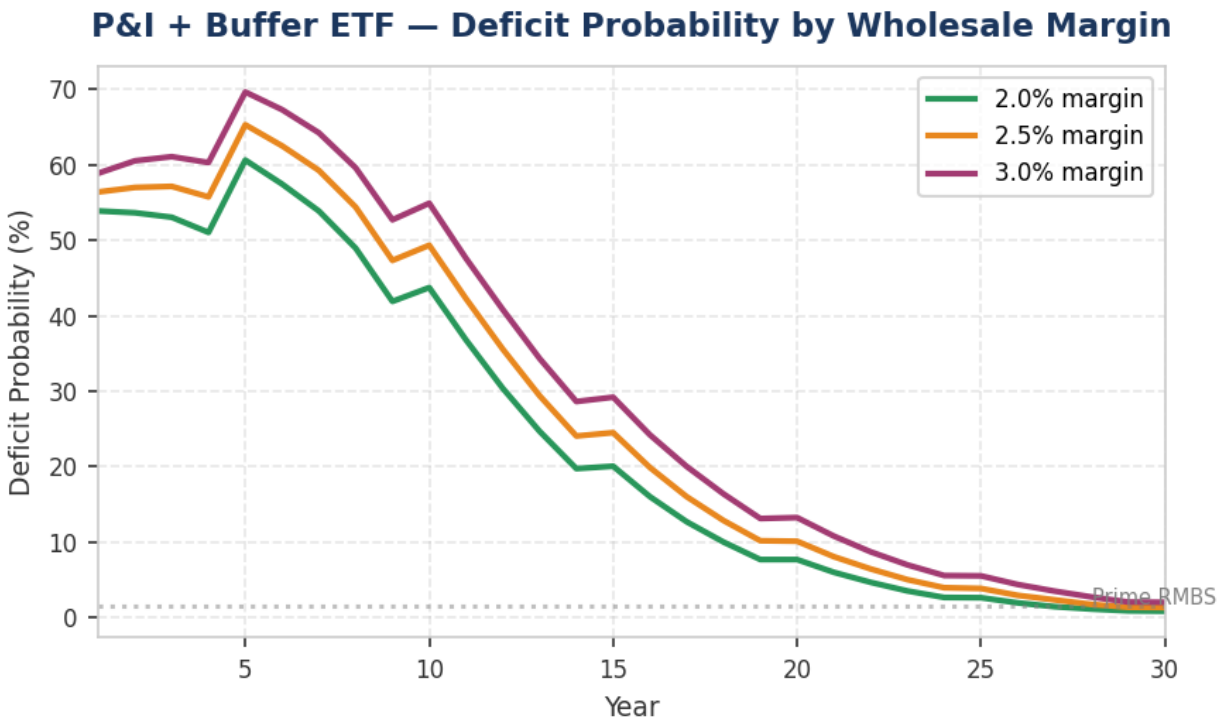
Product Overview for Funders

Feature	Detail	Funder Relevance
Collateral	Residential property (\$2M+)	Prime residential, first mortgage
LVR	60-80%	Conservative LVR range
Loan Type	Principal & Interest (recommended)	Self-amortising, no balloon
Loan Size	\$1.0-1.5M typical	Institutional scale
Tenure	Up to 30 years	Long-duration, stable AUM
Borrower Profile	Age 55-75, high-equity homeowners	Low-risk demographic
Investment	BlackRock Buffer ETF	Institutional fund management
Interest Coverage	Paid from investment returns	Not reliant on borrower income

How the Funder Gets Paid

- Wholesale margin applied above the cash rate — paid quarterly from the equity investment portfolio
- During market holidays, interest is deferred (capitalised) — NOT defaulted
- P&I structure means loan balance is continuously reducing
- At exit: loan repaid from investment portfolio and/or property sale

Credit Risk Analysis



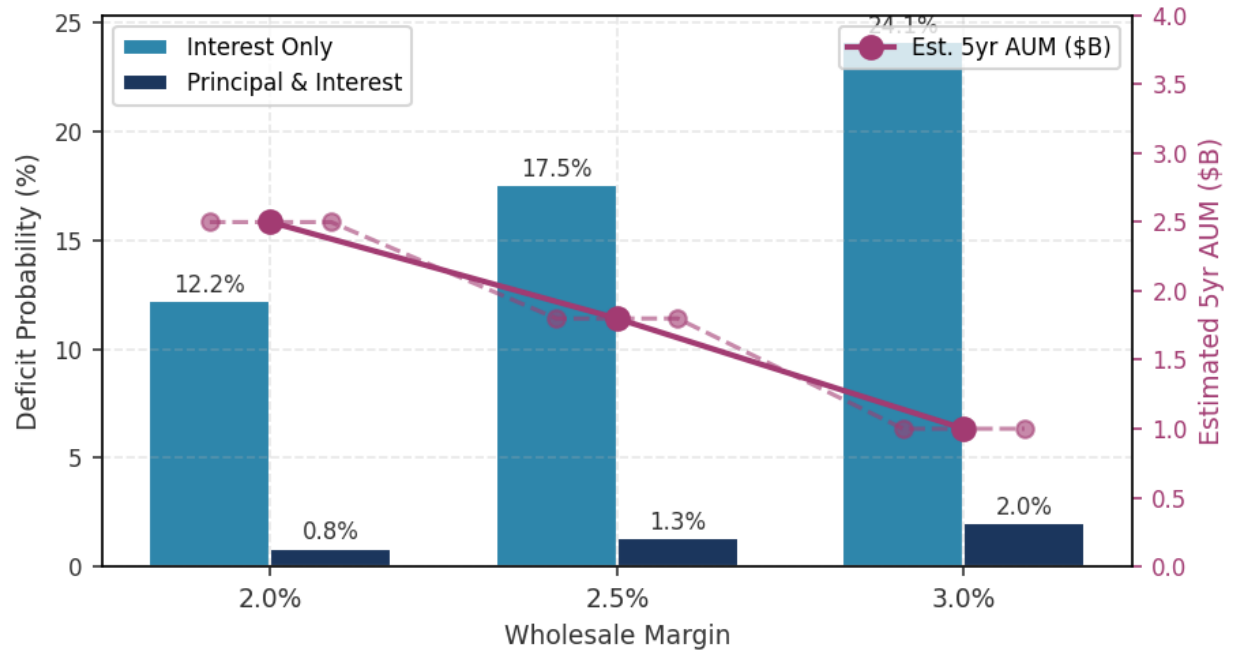
Key Credit Metrics (P&I + Buffer ETF, Optimised Structure)

Wholesale Margin	Deficit Probability	Expected Loss Given Claim	Fair Premium	Mean Surplus
2.0%	0.8%	\$332,263	\$772	\$3,781,357
2.5%	1.3%	\$362,673	\$1,327	\$3,488,771
3.0%	2.0%	\$394,903	\$2,202	\$3,202,495

Prime Australian residential mortgage default rates are typically 0.5-1.5% over the life of the loan. The EPM P&I structure achieves 2.0% deficit even at 3.0% wholesale margin — comparable to or better than prime residential mortgage loss rates.

Why Lower Margin = Better Product = Bigger Market

Wholesale Margin Impact — Deficit Probability & Market Size



The Margin-Volume Trade-off

Margin	P&I Deficit	IO Deficit	IO Viable?	Market Impact
1.5%	0.5%	7.9%	■ Yes	Both tiers available, maximum market
2.0%	0.8%	12.2%	■ Marginal	P&I dominant, IO premium tier limited
2.5%	1.3%	17.5%	■ No	P&I only, reduced market
3.0%	2.0%	24.1%	■ No	P&I only, smallest market

The Business Case for Competitive Pricing

- At 1.5%: Both IO and P&I tiers viable → estimated 2,000+ EPMS within 5 years → \$2.5B+ AUM
- At 3.0%: P&I only viable → estimated 800 EPMS within 5 years → \$1.0B AUM
- Lower margin per EPM × more EPMS = higher total revenue for the wholesale funder
- Long-duration assets (15-30yr mean life) provide stable, predictable fee income
- P&I amortisation means credit risk improves every year as LVR declines

Collateral & Security Structure

First Mortgage Security

- Registered first mortgage over residential property
- Minimum property value: \$1.5M (typically \$2M+)
- Maximum LVR at origination: 80%
- Property independently valued at origination

Investment Portfolio as Additional Security

- Loan funds are invested in BlackRock-managed Buffer ETF
- Investment portfolio provides secondary collateral
- In the P&I recommended structure, the loan amortises while the investment compounds
- By year 15: typical LVR has declined to ~30% (loan reduced, property appreciated)

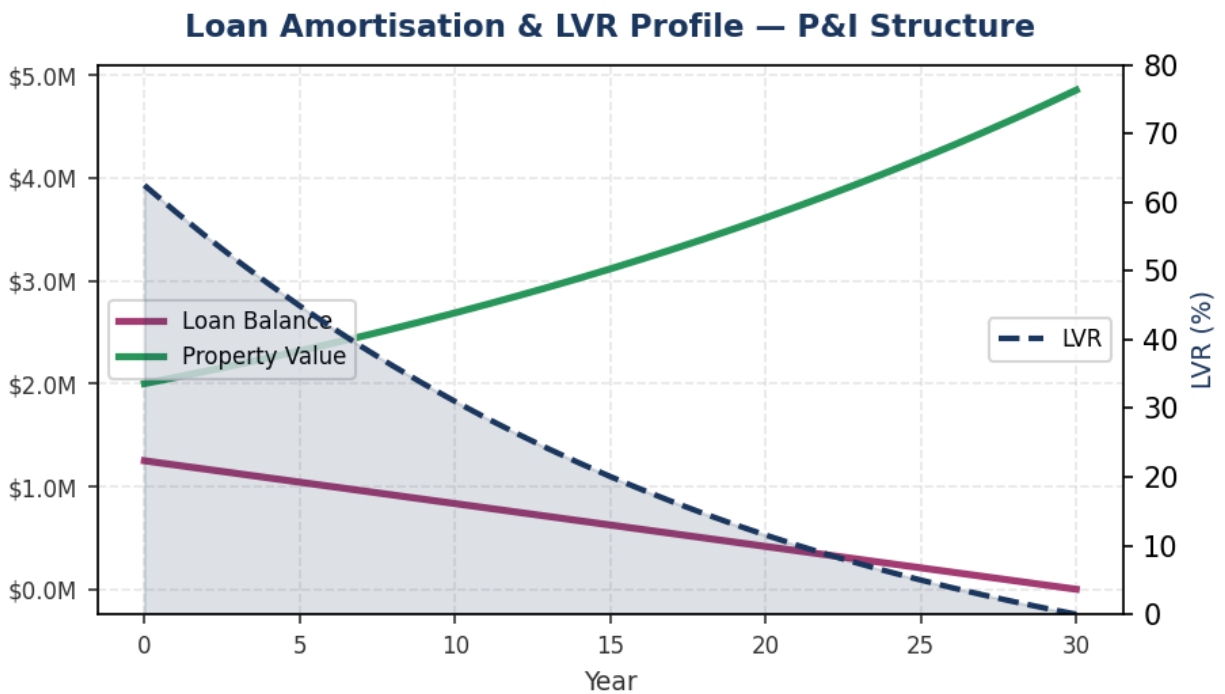
Deficit Protection (Insurance)

- FutureProof maintains insurance reserve funded by fair premium at origination
- Reinsurance cedes 20% of premium to third-party capital markets
- Insurance covers the gap if investment portfolio < loan balance at exit
- With P&I + buffer: insurance is called in only 0.5% of scenarios

Waterfall at Exit

1. Investment portfolio liquidated
2. Loan balance repaid to wholesale funder (priority claim)
3. Any deferred interest repaid
4. Surplus returned to homeowner/estate
5. If investment < loan: insurance covers deficit; if insurance insufficient: property sale proceeds

Loan Amortisation & LVR Profile

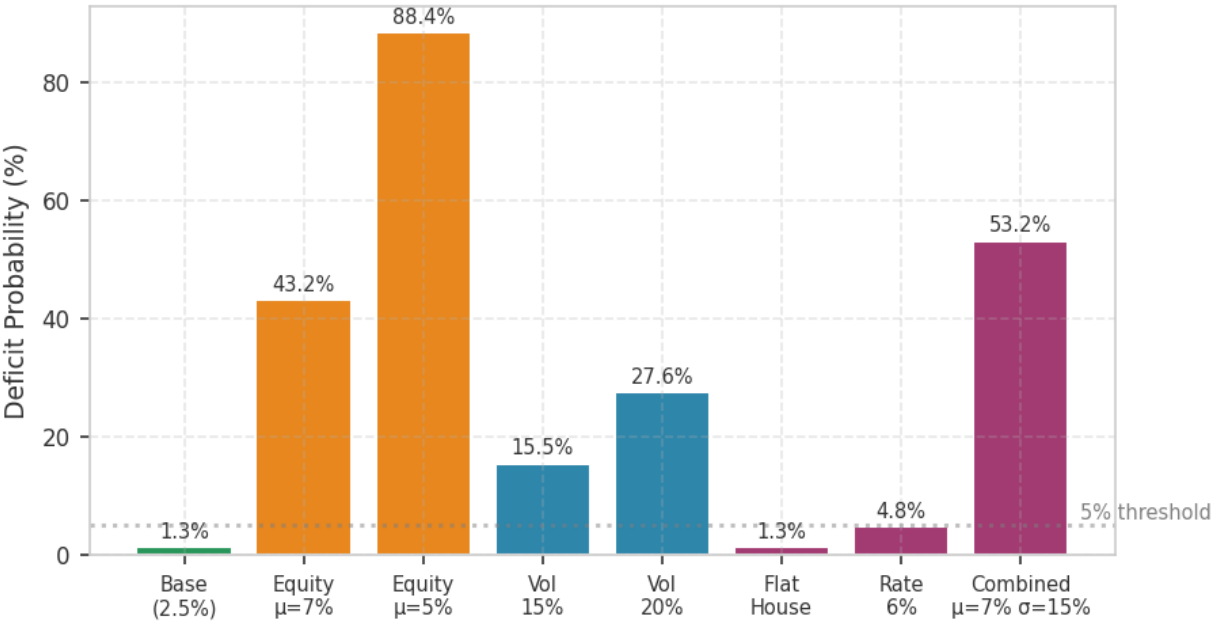


The chart above illustrates the declining funder exposure under the P&I structure. The loan balance amortises linearly to zero over 30 years while property values appreciate at ~3% p.a. The resulting LVR declines from 62.5% at origination to under 20% by year 15. This means the wholesale funder's security position improves every single year — even in a flat property market, the loan reduction alone drives LVR improvement.

Year	Loan Balance	Est. Property Value	LVR
0	\$1,250,000	\$2,000,000	62.5%
5	\$1,041,667	\$2,318,548	44.9%
10	\$833,333	\$2,687,833	31.0%
15	\$625,000	\$3,115,942	20.1%
20	\$416,667	\$3,612,222	11.5%
25	\$208,333	\$4,187,757	5.0%
30	\$0	\$4,854,978	0.0%

Stress Testing

Stress Testing — P&I + Buffer ETF at 2.5% Margin



Stress Scenario	Deficit Prob	Context
Base case (2.5% margin)	1.3%	Normal conditions
Lower equity return (7%)	43.2%	Prolonged low-return environment
Much lower return (5%)	88.4%	Severe underperformance
Higher volatility (15%)	15.5%	Crisis-level volatility
Extreme volatility (20%)	27.6%	GFC-type scenario
Flat house prices + vol	1.3%	Property market stress
Higher rates (6%)	4.8%	Rate tightening cycle
Combined ($\mu=7\%$, $\sigma=15\%$)	53.2%	Multiple adverse factors

Key message: Under normal conditions with a 2.5% margin, P&I + buffer ETF achieves just 1.3% deficit. Even under rate stress (6% cash rate), deficit remains at 4.8%. The product is most sensitive to equity return assumptions — but even at 7% equity return, the first mortgage security over \$2M+ property remains fully intact. The funder's exposure is to the loan balance, not the investment portfolio.

Portfolio Diversification Benefits

Why a Book of EPMs is Safer Than a Single EPM

- Different origination dates mean different equity market entry points
- Vintage diversification smooths returns across market cycles
- P&I amortisation means older EPMs have very low residual exposure
- Prepayment and mortality create natural portfolio runoff (mean exit year ~19)
- Insurance reserve pools risk across the entire book

Illustrative Portfolio Build-Up

Year	New EPMs	Total Active	Total AUM	Avg LVR
1	100	100	\$125M	62.5%
3	300	700	\$875M	55%
5	500	2,000	\$2.5B	48%
10	500	4,000	\$5.0B	40%

Note: Average LVR declines over time as P&I amortisation reduces loan balances and property values appreciate. A portfolio of seasoned EPMs has significantly lower credit risk than new originations.

The Partnership Proposition

What FutureProof Offers the Wholesale Funder

- Access to a new, growing asset class with ultra-low credit risk
- Long-duration, stable AUM (mean EPM life 15-20 years)
- Self-amortising P&I loans — credit quality improves every year
- First mortgage security over prime residential property (\$2M+)
- Institutional fund management (BlackRock) removes counterparty risk on investments
- Insurance layer provides additional loss protection

What FutureProof Seeks

- Competitive wholesale margin (target: 1.5-2.0% above cash rate)
- Commitment to fund an initial pilot of 50-100 EPMs (\$62-125M)
- Scalable facility to support growth to \$2.5B+ over 5 years
- Collaborative approach to product development and risk sharing

The Win-Win

- Lower margin → viable IO tier → larger addressable market → more AUM for the funder
- At 2.0% margin on \$2.5B portfolio = \$50M annual wholesale revenue
- At 3.0% on a smaller \$1.0B portfolio = \$30M — lower total revenue despite higher per-unit margin
- Long-duration, low-risk, self-amortising residential assets are exactly what bank balance sheets want

Methodology

Component	Detail
Simulation engine	50,000-path Monte Carlo, quarterly time steps, 30-year tenure
Risk factors	3-factor correlated stochastic model (equity, rates, house prices)
Correlation	Cholesky decomposition for correlated risk factors
Equity assumptions	10% mean return, 10% volatility (buffer ETF reduces to 8%)
Exit modelling	Gompertz mortality and constant-hazard prepayment
Implementation	Rust, 600K+ paths/sec, fair premium SE < 1%
Validation	Independent — not produced by FutureProof or the fund manager